

Durability-Driven Bond Degradation Mechanism of GFRP/HFRP Rock Anchors under Coupled Groundwater and Sustained Loading Conditions

1. Objective

To investigate the durability and bond behavior of GFRP and HFRP bars embedded in grout under combined environmental conditioning and sustained loading.

2. Materials Selection

- a. *Two types of FRP bars are selected:*
 - i. GFRP bars having diameter of around 13 mm (Timur_25) and 70% fiber volume.
 - ii. HFRP bars (Hybrid FRP): Uniformly dispersed Hybrid FRP (UDH) has been reported to have better durability against moisture and also performs well in terms of its pseudo-ductile behavior. However, the fatigue behavior of shell type HFRP is relatively better. The ratio of carbon fiber will be 40% for positive hybridization effect.
- b. *Grout that will surround the bonded length of the FRP. (Mention standard here)*
- c. *Concrete (Mention standard here)*

3. Environmental Conditioning:

- a. *Environmental conditioning of bar pull-out specimens with sustained load.*

There are various ways in which this test has been conducted.

 1. Apply environmental exposure without sustained load, apply sustained load separately without exposure and then compare results. (Gao Rui Thesis)
 2. Apply environmental exposure without sustained load. Then apply sustained load is applied as the next step. We will use this method for transverse shear test. This method involved passing a bar through three concrete blocks joined together on thus defining a predetermined joint plane. Then after conditioning at environmental exposure, shear load can be applied. This test will be performed in 2nd phase of research and although the exact test setup has not been defined, there are some references of similar setup. (Guan_2025)

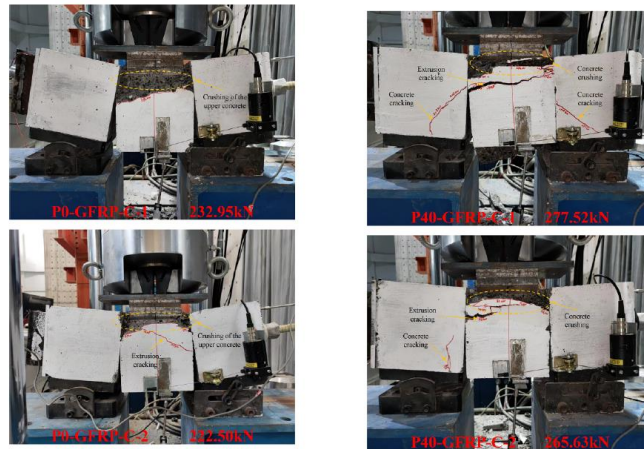
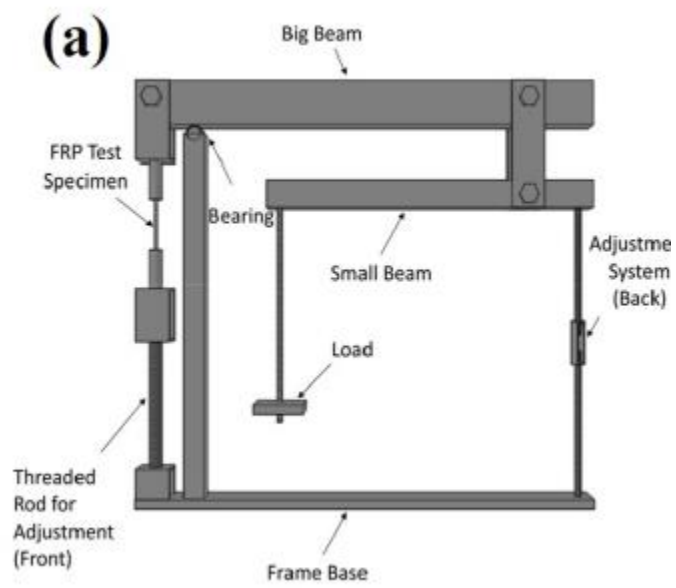


Figure 1 Transverse shear test on Concrete with restressed FRP bar passed (Gian_2023)

3. Apply environmental conditioning and load simultaneously. (Li Chen_25, Tang_22) We will use this method for bond durability test using Creep frame





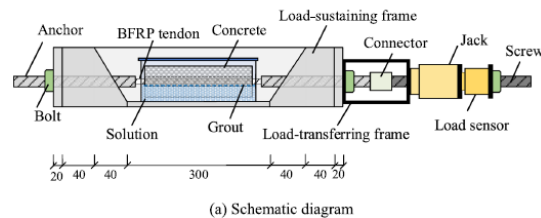
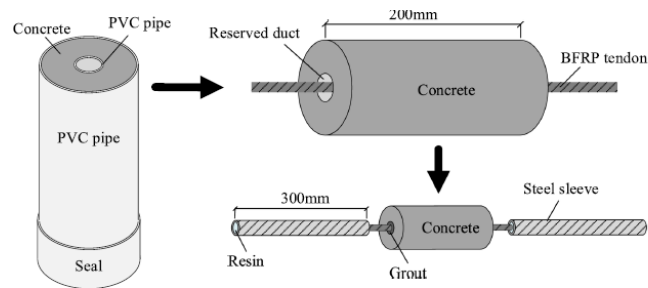
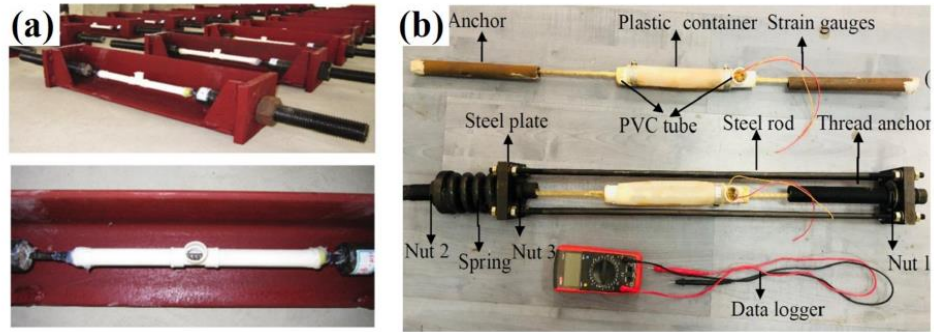
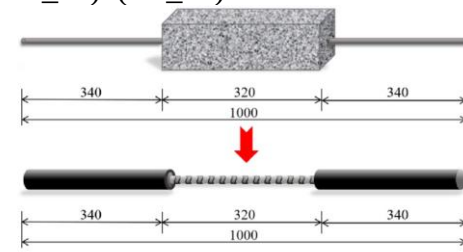


Figure 2 Shi_25

- b. *Environmental conditioning of pull-out specimens without sustained load*
For pull out specimens under 0 sustained load do not require a creep frame and can be place in solution tank directly. For this we would require three tanks at different 25C, 40C and 60C temperature. The size of each tank should roughly be 1.2m x 1m x 0.8 m. This should be provided with temperature regulation device for which various methods can be adopted.
- c. *Environmental conditioning of GFRP / HFRP bars:* In order to evaluate the performance of bars alone, we also need to perform durability test on bars. Indirect method of immersion is most preferred method based on literature as the surrounding grout can trigger a realistic alkaline environment. (Feng_22,

Pan_25). (Xu_24)



(a)



(b)

Figure 3 Xu_2024 70mmx70mmx320mm

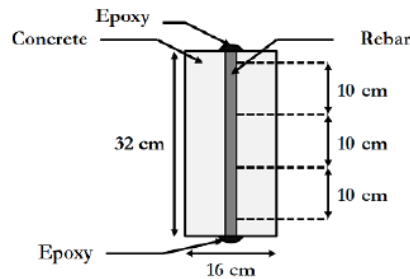


Figure 4 Indirect Immersion (Rolland_21)

d. *Environmental conditioning of grout and concrete*

This required for determining the influence of environmental expose alone on the grout. The size of cube can be 100mm x 100mm x 100mm (Li_24)

4. Material Specification:

- a. *Pull out Test:* Sample size of 800 mm with portion of bar (up to 200 mm) embedded in grout. This will act as first environment between the surface of bar and the grout. The length of the bond can be 4D-6D to engage the bond fully and enable pull out failure of the bond. The rest of the length inside the bar can be covered by PVC sleeve that can act as 'bond breaker'
For the second environment (grout-rock interface), some researchers have used granite cube inside which hole is drilled for grout. Concrete block can also be used. This provides realistic confinement for pull out test, however, there is ambiguity regarding whether or not a 2nd environment can be applied during the environmental conditioning due to constraints of the creep frame to be used for sustained loading. In case it is not possible we might not add 2nd environment. For confinement during pull out test we can add rings around the grout during put out testing or perform it on specialized equipment such as

Modified Hoek call (Li_24) . 5 specimens will be tested before the conditioning and 3 specimens after conditioning.

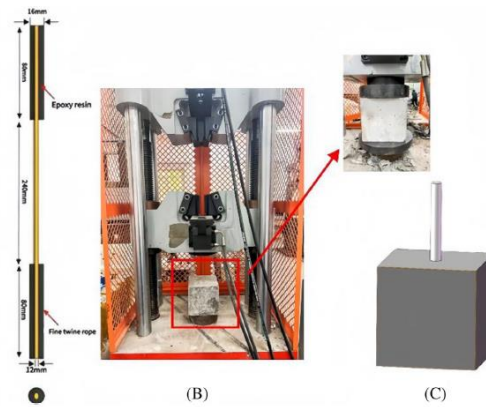


Figure 5 Ren_24 Pull out assembly (mortar used as grout)

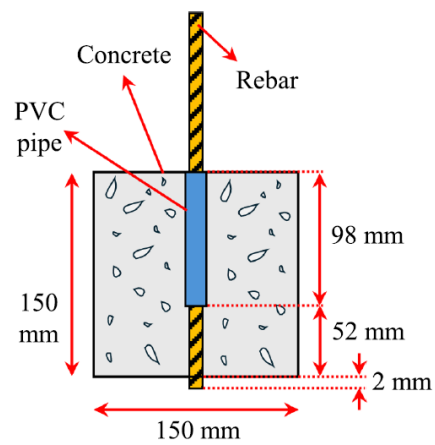


Figure 6 Timur_25 no grout was used (CAN/CSA-S805-12 Pull out test standard)

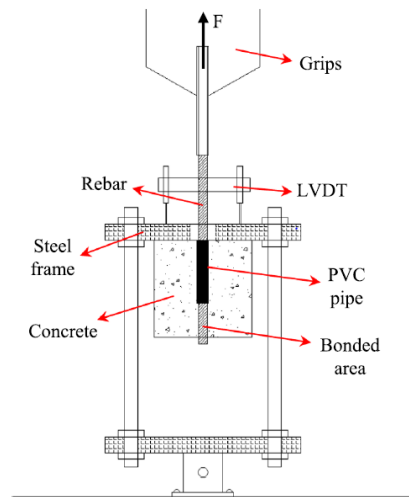


Figure 7 Set up for Pull out test Timur_25

- b. *Tensile Test of Bar*: Tensile test of bar before and after conditioning. 600-800 mm bar specimens will be used for this purpose. Steel sleeves (JSCE-E531) covering 150-300 mm at both end will be added filled with grout to prevent premature failure of the bar. The remaining 200-300 mm (40D as per ACI 440.3R-12) will be open length.

This test can be performed as per ASTM D7205 and will require 5 specimens before environmental conditioning and 3 specimens after each exposure condition. Additionally, basic tests for determining the material properties before and after exposure such as diameter, Elastic modulus etc. will also be determined.

- c. *Short Beam Test*: This test is performed for Inter-laminar shear strength of the bar. Based on literature study literature, this test is required in case of bond durability tests as after environmental conditioning, the ILSS occurs in during the bar pull out tests.

This test can be performed as per ASTM D44750-02 standard. The length of 6D of which 5D is span. 5 specimens will be needed. The loading rate will be 1.3mm/min.

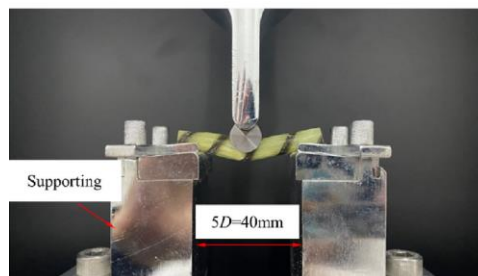


Figure 8 Short beam shear test (Pan_25)

- d. *Transverse Shear test*: Performed as per ASTM 7617. The sample size of 225mm is needed. The loading rate of 1.3mm/min. The equation ($P/2A$) can be used (Xu_2024)
- e. *Compressive Strength of grout*: Before and after environmental exposure.
- f. *Compressive Strength of Concrete*: Before and after environmental exposure
- g. *Moisture Absorption Test*: Before and after environmental exposure. Test can be performed as per procedure adopt in Thesis by Guo Rui. It used a sample size of 240 mm which is placed in a oven at 60C for 7 days. Then it can be placed in the solution at different temperatures. For this test, 5 samples are required.
- h. *Microscopic Analysis*: Before and after environmental exposure. SEM (10mm x10mm x 2mm), X-Ray CT, DSC, FTIR etc. can be conducted

5. Bond Test Configuration

Most of the literature studied focuses of the durability of bars rather than the bond. In case of bond durability most papers consider bars embedded in concrete with no surrounding grout. Based on literature some methods than can be adopted are as follow

- a. Configuration 1: Provide the grout (environment 1) as well as concrete/rock environment (environment 2). This would require 100mm concrete cube with hole at the center filled with grout and where FRP can be inserted. This closely represents field anchor conditions. However, this setup is difficult to integrate with creep frame and solution chamber.
- b. Grout only specimen with no concrete confinement. This setup can be easily applied to the creep fame during sustained load. However, the pull out test is not very realistic as it lacks the radial confinement. Confinement can later be applied during pull out test using
 - i. Sleeve rings
 - ii. Specialized apparatus such as modified Hoek Cell (Triaxial)(Li_24)
 - iii. The specimen after exposure surrounding the grout can later be placed in fresh concrete specimen and allowed to cure for 28 days. There is no literature for this method is only my personal idea.

6. Test Variables

- a. *Bar Type*: GFRP and HFRP as discussed earlier
- b. *Environment type*: The environment I chose is Ground Water which can be acidic, alkaline or neutral. I have only found one article from Australia (Matthew_2018) which uses Ground water solution for environmental exposure. The pH of alkaline ground water varies between 7-10 and that of acidic nature has pH below 4. Chinese standards in particular regarding the particular ionic composition of acidic or alkaline ground water
 - i. Alkaline: Sodium bicarbonate + NaOH + NaCl
 - ii. Acidic: Sodium hydrogen sulfate + NaCl

However, there are other options for environmental solutions across literature.

- i. Combined Freeze Thaw and Alkaline Environment (Ren_24)

- ii. Acid Mine Drainage (Ginish_25)
- iii. Medium (ph=12.5) to high alkalinity (pH=13.4) (Feng_22)
- iv. Wet Dry Cycles (Randel_21)
- v. Hygrothermal (Guo Rui Thesis)

Feedback from supervisor regarding suitable environmental exposure selection is required.

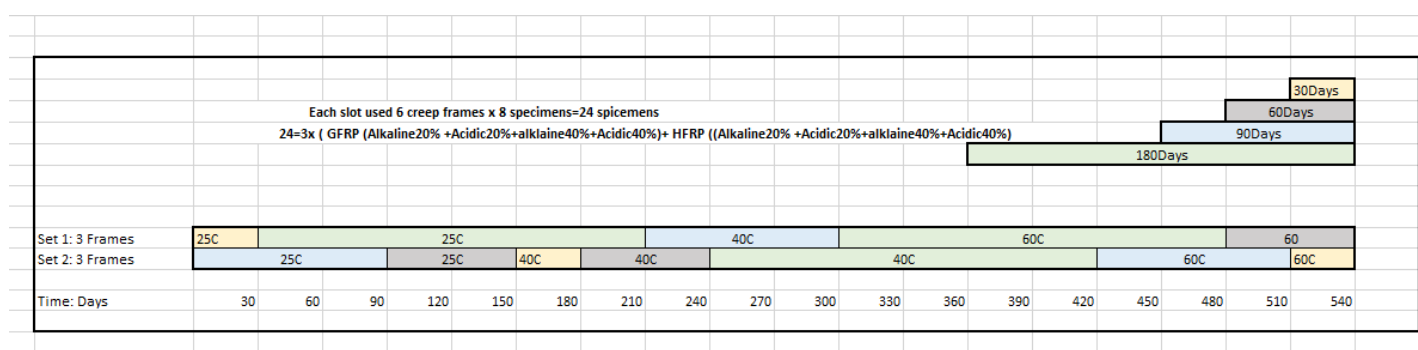
- c. *Temperature*: Most of the literature uses three temperature ranges. 22-30 C as lower temperature, 40-45C as medium temperature and 55-80C as high temperature. (Xu-24, Feng_22, Rolland_21, Pan_25, Shi_25). The upper temperature range should be lower than the glass transition temperature. For our research three temperature ranges (25C, 40C , 60C) seem suitable as they fall within the range.
 - d. *Time*: Different durations have been used across the research for different environmental conditions. However, conditioning for up to 6 months is most common. The time interval selected are 0 days, 30 days, 60 days, 90 days and 180 days. Other instances of time across literature are: 7-31-62-90-126 Days (Migan-24) for Saline solution, 0-25-80 days (Ren_24) for freeze thaw, 30-60-90—180 days (Wang_23), 417 days (Feng) for creep, 240 days (Rolland), and 21, 40 and 60 days (Pan_25).
 - e. *Sustained load*: This caused micro cracks and creep which can accelerate the bond degradation. For our research 3 ranges of sustained load have been taken (0%-20%-40% of the Ultimate Tensile load and are in line with the research read. (Tang_22, Li Chen_25)
7. **Specimen Distribution:**
Given below is the estimated number of specimens of HFRP and GFRP for testing. In addition to that ingredient for grout and concrete will also be required which will be estimated based on W/C ratio, required strength and rest of the parameters.

Table 1 Number of Specimens required for each test

Test Name	Bar Type	Environment Type	Temperature	Time	Sustained load	Specimens/ condition	Length of Specimen	Total Specimens
Tensile Test of Bar	GFRP , HFRP	Alkaline GW, Acidic GW	25C 40C 60C	0, 30, 60,90,180 days	0% Only	3	800mm	154 specimens
Short Beam Test	GFRP , HFRP	Alkaline GW, Acidic GW	25C 40C 60C	0, 30, 60,90,180 days	0% Only	5	80mm	250 specimens
Moisture Absorption	GFRP , HFRP	Alkaline GW, Acidic GW	25C 40C 60C	0, 30, 60,90,180 days	0% Only	5	240mm	250 specimens
Pull Out test	GFRP , HFRP	Alkaline GW, Acidic GW	25C 40C 60C	0, 30, 60,90,180 days	0% , 20%, 40%	3	800mm	442 specimens

Transverse Shear test on bar	GFRP , HFRP	Alkaline GW, Acidic GW	25C 40C 60C	0, 30, 60,90,180 days	0% Only	5	225mm	250 specimens
Transverse Shear on Bar embedded in grout and surrounded by concrete	GFRP , HFRP	Alkaline GW, Acidic GW	25C 40C 60C	0, 30, 60,90,180 days	0% Only	5	225mm	250 specimens

- 8. Time line for Environmental exposure of Bars embedded in grout under sustained loads:** In this research the most critical and time-consuming aspect is the environmental testing under sustained loading. Currently three creep frames are available, however, timeline has been shared based on 6 loading frames, which means 3 additional creep frames need to be purchased. The time line has been scheduled keeping the following aspects in mind
- Each frame can have 8 specimens simultaneously.
 - Each frame can have one temperature for all specimens.
 - Based on the above two 6 frames can be divided into two 2 sets, each having 24 specimens on same temperature. For example, frame 1, frame 2 and frame 3 for 25 C and frame 4, frame 5 and frame 6 for 40 C
 - Each frame of 8 specimens can have different solution types. The 24 specimens for each set have been divided as: 3 x (Bar Type (2) x (Solution Type (2) x Sustained load level (2)) = 24. Accordingly, the schedule has been set and takes approximately 1.5 years for these tests to complete. The schedule is shared below



During this duration the other tests such as Environmental exposure of bars Embedded in grout under 0% sustained loads, bare bars, tensile tests, short beam test etc. will also be performed

9. Equipment Required:

- 6 Creep frames (3 available and 3 needs to be purchased) with the required accessorized such as load device, displacement measure device, solution cell, loading frame, connector, temperature regulator etc.

- b. 6 Polypropylene tanks with temperature regulation. (Dimensions 1.2m x 1 m x 0.8m). Temperature sensor and pH meter may also be required.
- c. Equipment for pull-out test. This also required strain measurement apparatus for slip measurement.
- d. Required apparatus for grout/concrete mixing and all related tests related to concrete/ grout preparation

10. Critical decisions needing supervisors Input:

- a. Bond test method (Concrete cube vs grout only)
- b. Solution Selection
- c. Feasibility of creep frame immersion system
- d. Number of frames